



Ensemble selection with joint spectral clustering and structural sparsity^{*}

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ARTICLE INFO

Article history:

Received 19 October 2020

Revised 16 April 2021

Accepted 19 May 2021

Available online 29 May 2021

Keywords:

Ensemble selection

Structural sparsity

Unsupervised selection

Spectral clustering

Dynamic and static

Robustness

ABSTRACT

Generally, ensemble selection techniques are split into two categories: dynamic and static. Static ensemble selection selects a fixed subset of the original ensemble which improves the space complexity but is not flexible to each test instance. Dynamic ensemble selection selects base learners on-the-fly according to each test instance but it does not significantly improve the complexity. Currently, there is no ensemble selection technique that is robust to the test instances as well as improves space complexity. To narrow this gap, we propose a novel static ensemble selection method, called Ensemble Selection with Joint Spectral Clustering and Structural Sparsity. This method integrates spectral clustering and structural sparsity into a joint framework whose ensemble selection result is robust to test instances and consumes less space. Using 25 datasets from KEEL and UCI, we demonstrate the effectiveness of our proposed algorithm and its promising performance compared to that of other state-of-the-art algorithms.

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1. Introduction

Ensemble learning, also known as committee-based learning or learning multiple classifier systems, aims to combine a number of learners called base learners to achieve generalization performance superior to that of individual learner [1,2]. Typically, the ensemble can achieve the state-of-the-art prediction performance [3,4]. Because reducing the number of base learners can improve the ensemble's accuracy and reduce its complexity, ensemble selection, sometimes called ensemble pruning, is often adopted to select only a subset of base learners from an ensemble to obtain a better performance with less cost [1].

Generally, ensemble selection techniques are split into two categories: dynamic and static [5]. Static ensemble selection selects a fixed subset of the original ensemble for all test instances. Dif-

ferent criteria have been designed to select the best base learners in static ensemble selection to improve the ensemble learning performance. Ordering-based static selection methods order individual base learners through a specific criterion, after which the top k base learners are selected [6,7]. Optimization-based selection methods formulate ensemble selection as an optimization problem that aims to determine the best base learners [8–10]. Naturally, static ensemble selection considers two goals simultaneously: maximizing the generalization performance and minimizing the number of learners [1,9]. Therefore, static ensemble selection improves the space complexity as it simply retains a small percentage of the original ensemble. However, a clear disadvantage of static ensemble selection is that a fixed subset is not flexible to every test instance, which may decrease the performance.

In dynamic ensemble selection, the base learners are selected on-the-fly through a specific evaluation criterion according to each test instance to be classified [11,12]. Typically, to select the best base learners, competence of each is estimated based on a local region of the feature space where the test instance is located [13,14]. The rationale behind these approaches is that no individual in an ensemble is an expert in predicting all test instances; rather, each base learner is an expert in a different local region of the feature space [15]. A clear limitation of dynamic ensemble selection is that the space complexity is similar to that of the original ensemble. Currently, there is no ensemble selection technique that is both robust to base learners as well as improves space complexity.

^{*} This work is supported by the National Key Research and Development Plan (2018YFB1004401), NSFC under the grant No. 61732006, 61772536, 61772537, 62072460, 62076245 and Beijing Natural Science Foundation under the grant No. 4212022. This study was also partially supported by the key R&D program of science and technology foundation of Hebei Province (19210310D). Hebei Key Laboratory of Machine Learning and Computational Intelligence, College of Mathematics and Information Science, Hebei University, Baoding, China.

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Both static and dynamic ensemble selection typically conduct selection through supervised learning, the class labels in the validation data supervise the selection of base learners. Therefore, such ensemble selection results are sensitive to the local prediction values of the base learners. For example, one base learner's incorrect prediction on specific instances may affect the ensemble selection result. In fact, the original data can be transformed into an unlabeled prediction space by the ensemble. The method of transformation will be presented in Section 3.2. In the prediction space, the correct prediction power of each base learner is leveraged, after which the global prediction distribution of the ensemble is described. The selective ensemble based on such a prediction space is no longer sensitive to the local prediction results of each base learner. Subsequently, the robust ensemble selection problem is transformed into the problem of selecting the most competent base learners from the unlabeled transformed data (i.e., the prediction space).

Unsupervised ensemble selection, similar to unsupervised feature selection [16], aims to select discriminative base learners by maintaining the information or structure of the prediction space as invariant. The applications of graph spectral theory in unsupervised feature selection have proven to be strong for dimensionality reduction [16]. Inspired by the recent developments in spectral analysis and sparse coding for unsupervised feature selection [17,18], we propose a robust static ensemble selection method called Ensemble Selection with Joint Spectral Clustering and Structural Sparsity (ES-JSS). Specifically, we select competent base learners so that the global prediction distribution of the data can be well preserved. To select the most competent base learners, spectral clustering is proposed to learn pseudo cluster-labels on the transformed data simultaneously in the procedure of ensemble selection. The learned pseudo cluster-labels closely reflect the global prediction distribution and preserve the correlations between different base learners. Meanwhile, the competent base learners are weighted using structural analysis with $l_{2,1}$ -norm regularization to maintain the global prediction distribution (i.e., the pseudo cluster-labels) as invariant. Thus, the most competent base learners are correlated and selected as they jointly reflect the global prediction distribution.

Unsupervised ensemble selection is conducted on the global structure of prediction space. Then, the small perturbation of prediction values of each base learner can be neglected by unsupervised learning. Thus, this unsupervised learning is robust not only to test instances, but also to the type of base learners. However, it is worth noting that only the base learner with [0,1] interval prediction is suitable for our proposed unsupervised method; otherwise, data transformation does not work and global prediction distribution space can not be achieved.

The main contributions of this study can be summarized as follows:

- A prediction space, which leverages the prediction power of an ensemble, serves as the basis of ensemble selection. The data in the prediction space are unlabeled, but they globally describe the prediction distribution of the ensemble.
- To the best of our knowledge, this is the first time to adopt unsupervised techniques for selective ensemble learning by simultaneously integrating clustering and sparsity.
- Experimental results demonstrate that our proposed static ensemble selection method ES-JSS is robust to test instances (i.e., it achieves better prediction performance with less space complexity in most cases).

The remainder of this paper is organized as follows: Section 2 briefly reviews some closely related works and introduces the concepts of structural sparsity and spectral clustering. In Section 3, we present our proposed framework with joint

spectral clustering and structural sparsity in the prediction space. Section 4 presents our ES-JSS optimization algorithm and analyzes its convergence. Section 5 reports on extensive comparison experiments. Finally, the paper is concluded in Section 6.

2. Preliminaries

2.1. Ensemble selection

Static ensemble selection has two branches: ordering-based selection and optimization-based selection. Ordering-based methods use different measures to order the base learners. For example, Reduce-Error Pruning (RE) [6] selects the base learners by estimating the classification error. Kappa Pruning (Kappa) [6] selects the most diverse base learners by using Kappa statistic k_p . Complementariness Pruning (CP) [7] incorporates the base learner whose performance is most complementary to the selected subensemble at each iteration. Margin Distance Pruning (MD) [7] defines a signature vector and aims to find a subensemble whose average signature vector is closest to a reference position. Furthermore, some optimization-based methods formulate ensemble selection as an optimization problem to find the most competent base learners. For example, Diversity Regularized Ensemble Pruning (DREP) [8] employs an explicit diversity regularization to ensemble selection. Pareto Ensemble Pruning (PEP) [9] proposes a bi-objective formulation method to maximize the generalization performance while minimizing the number of base learners. Moreover, Distributed Objection Maximization for Ensemble Pruning (DOME) [10] formalizes ensemble selection as an objection maximization problem with a general distributed framework. The method proposed in [19], as a newest static ensemble selection algorithm, prunes an ensemble using a novel optimization model by a difference of convex algorithm (DCA). In summary, these static methods share a common characteristic that a fixed selection result is obtained; they improve the space complexity, but they are not flexible to every test instance, which may degrade their performance on some datasets.

Many dynamic ensemble selection methods have been proposed to adaptively select the most competent base learners available in an ensemble to improve the predictive performance for every test instance [11,12]. Typically, the competence of a base learner is estimated based on a local region of the feature space where the test instance is located [13,14]. For example, both K -Nearest Oracles Eliminate (KNORA-E) [20] and K -Nearest Oracles Union (KNORA-U) [20] use K -nearest neighbours (KNN) to find the neighborhood of the test instance that represents the local region of competence and then they select the most competent base learners based on this region. KNOP is similar to KNORA-U but is defined in the decision space [12]. Fire-DES++ [21] is a newly proposed dynamic ensemble selection framework that reduces the overlap of classes in the validation set, and defines the region of competence using an equal number of samples in each class. Meanwhile, some other methods focus on another key problem (i.e., how to estimate the competence level of each base learner). Randomized Reference Classifier (RRC) [14], Kullback-Leibler (DES-KL) [22], and dynamic Ensemble Selection Performance (DES-P) [22] adopt Probabilistic selection criteria; DES-MI relies on a weighting mechanism to highlight the competence of base learners that accurately classify samples in the region of underrepresented competence [13]. META-DES [23] is designed based on Meta-Learning to evaluate whether it is competent enough to classify a query sample. However, it should be noted that a clear limitation of dynamic ensemble selection is that the space complexity is similar to that of the original ensemble.

Moverover, some other methods consider ensemble selection on some weak supervised problems such as imbalanced data [24] and

multi-label data [25]. Further, some methods focus on regression ensemble selection [26]. To the best of our knowledge, all ensemble selection methods are supervised. To date no work has considered unsupervised ensemble selection.

2.2. Unsupervised feature selection

In recent years, unsupervised feature selection, which is able to identify and select relevant features without needing class label information [27], has raised considerable interest in many research areas. These works can be roughly divided into three groups: filter, wrapper and hybrid.

Filter unsupervised feature selection methods select the most relevant features through the data itself, i.e., features are evaluated based on intrinsic properties of the data [27]. The main goal of filter feature selection is to get a better result of feature selection. Some feature selection methods based on spectral analysis combined with sparse learning, have been proposed in the last years. For example, by a sparse optimization model with $l_{2,1}$ -norm, Structural Optimal Graph Feature Selection (SOGFS) is proposed, which simultaneously performs feature selection and local manifold structure learning [40]. Another method called Clustering Guided Sparse Structural Learning (CGSSL) conducts feature selection by jointly exploiting non-negative spectral analysis and clustering structural learning with sparsity [17]. All these filter methods focus on feature selection with the assistance of learning the cluster structure of the data.

Wrapper Unsupervised Feature Selection uses the results of clustering to evaluate feature subsets [27]. Methods developed under this approach usually select those features that contribute to get a comparable or more accurate clustering solution in high dimensions [27]. In well-studied field of wrapper unsupervised learning, many methods have introduced sparsity for feature weighting in clustering. For example, Sparse k-means clustering for feature selection is proposed by means of inducing l_1 and l_2 constraints into their clustering objective function [35]. Recently, Entropy Weighted Power k-means and Entropy Weighted Dirichlet Process means methods are proposed for feature weighting and determining the number of clusters simultaneously [38,39]. Furthermore, a lasso weighted k-mean algorithm (LW k-means) is proposed by imposing an l_1 regularization term to induce feature selection in a sparse clustering framework [37]. This is a state-of-the-art sparse learning based clustering method for high-dimension data. Unlike those k-mean type of clustering methods, [36] proposed a new clustering procedure by Influential Features PCA. All these wrapper methods focus on clustering simultaneously weighting features in high-dimensional data. Their main goal is to get a perfect cluster results in high dimensions.

The objective of hybrid unsupervised feature selection is to achieve both robust clustering and robust feature selection. To address the problem with outliers or noise, a hybrid method called Robust Unsupervised Feature Selection (RUFS) is proposed to perform feature selection while the pseudo labels are learned via local learning regularized robust nonnegative matrix factorization [18]. Following this idea, Robust Unsupervised Feature Selection via Matrix Factorization (RUFMS) is proposed, which simultaneously performs discriminative feature selection and robust clustering by using $l_{2,1}$ -norm [41]. Note that the qualities of both clustering and feature selection are considered as the evaluation criteria of hybrid methods.

In a word, sparsity and clustering have been successfully introduced in unsupervised feature selection. Inspired by this, we will conduct unsupervised ensemble selection by joint sparsity and clustering in this study.

2.3. Structural sparsity and spectral clustering

In this subsection, we briefly introduce the structural sparsity and spectral clustering elements of our proposed model.

Structural sparsity can not only realize the automatic selection of an ensemble but also makes the model easier to interpret. To obtain a row sparseness relation matrix, we adopt $l_{2,1}$ -norm as the regularization term of our objective function. Let $\mathbf{W} \in \mathbb{R}^{k \times c}$ represent a matrix whose i th row and (i, j) th element are denoted by $\mathbf{w}^i$ and W_{ij} , respectively. Then, the mathematical description of the $l_{2,1}$ -norm is as follows:

$$\|\mathbf{W}\|_{2,1} = \sum_i \sqrt{\sum_j W_{i,j}^2} = \sum_i \|\mathbf{w}^i\|_2. \quad (1)$$

In cluster analysis, graph-theoretic methods have been well-studied and utilized in many applications. As a graph-theoretic method, spectral clustering has been verified to be effective in detecting the cluster structures in data [17]. $\mathbf{E} \in \mathbb{R}^{n \times c}$ represents the pseudo cluster-label matrix, which describes the pseudo cluster-label distributions of n instances in $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n] \in \mathbb{R}^{m \times n}$, defined by Yang et al. [28] as follows:

$$\mathbf{E} = [f_1, f_2, \dots, f_c] = \mathbf{C}(\mathbf{C}^T \mathbf{C})^{-\frac{1}{2}}, \quad (2)$$

where $\mathbf{C} = [\mathbf{C}_1, \mathbf{C}_2, \dots, \mathbf{C}_c] \in \{0, 1\}^{n \times c}$ with $\mathbf{C}_i \in \{0, 1\}^{1 \times c}$ represents the pseudo cluster label vector for instance $\mathbf{x}_i$, and c represents the number of clusters. This means that $C_{ij} = 1$ if instance $\mathbf{x}_i$ is assigned to the j th cluster, and $C_{ij} = 0$ otherwise. From Eq. (1), we have

$$\mathbf{E}^T \mathbf{E} = (\mathbf{C}^T \mathbf{C})^{-\frac{1}{2}} \mathbf{C}^T \mathbf{C} (\mathbf{C}^T \mathbf{C})^{-\frac{1}{2}} = \mathbf{I}_c, \quad (3)$$

where $\mathbf{I}_c \in \mathbb{R}^{c \times c}$ represents an identity matrix.

The Ratio Cut method is widely used to evaluate the weights between subgraphs in spectral clustering learning. A Ratio Cut method of the spectral clustering model is expressed as follows:

$$\min_{\mathbf{E}^T \mathbf{E} = \mathbf{I}_c, \mathbf{E} \geq 0} \text{Tr}[\mathbf{E}^T \mathbf{L} \mathbf{E}]. \quad (4)$$

In Eq. (3), $\mathbf{L}$ denotes the Laplacian matrix and generally, $\mathbf{L} = \mathbf{A} - \mathbf{B}$, where $\mathbf{B} \in \mathbb{R}^{n \times n}$ denotes the similarity matrix on $\mathbf{X}$, and $\mathbf{A} \in \mathbb{R}^{n \times n}$ denotes a diagonal matrix with $A_{ii} = \sum_{j=1}^n B_{ij}$. Most spectral clustering models adopt the radial basis function (RBF) kernel to construct $\mathbf{B}$ as follows:

$$B_{ij} = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|_2^2}{2\sigma^2}\right). \quad (5)$$

Theoretically, $\mathbf{C}$ is a 0 or 1 matrix by definition. Unfortunately, the optimization of Eq. (3) is an NP-hard problem. Therefore, a commonly used strategy is to relax it to continuous values [18].

3. Our proposed framework

3.1. The notions

In this subsection, we summarize some notations used in the remainder of this paper. The input of a learning algorithm consists of a set of labeled instances $\mathbf{S} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^n$. Each instance is characterized by a feature vector $\mathbf{x}_i \in \mathbb{R}^m$, which is described by m feature values in a $m \times 1$ vector, and a class label $y_i \in \{1, 2, \dots, d\}$, which denotes the class label of the i th instance. $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n] \in \mathbb{R}^{m \times n}$ is an $m \times n$ matrix that represents the feature space of all instances. $\mathbf{Y} = (y_1, \dots, y_n) \in \{1, 2, \dots, d\}^{1 \times n}$ is a $1 \times n$ vector that represents the label space of all instances. The base learner h_i is a map: $\mathbf{X} \rightarrow \mathbf{Y}$ from feature space $\mathbf{X}$ onto label space $\mathbf{Y}$. Ensemble methods generate a set of base learners $H = \{h_i\}_{i=1}^k$, which are pooled to produce a final prediction.

For a given matrix $\mathbf{W} \in \mathbb{R}^{k \times c}$, $\mathbf{w}^i$ and W_{ij} denote the i th row and (i, j) th element, respectively. The trace, transpose, inverse, and

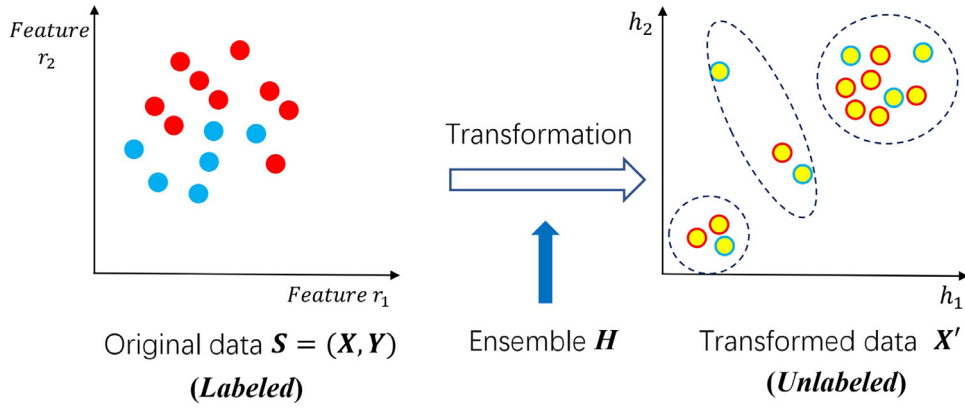


Fig. 1. Illustration of data transformation. r_1 and r_2 denote the features of the original labeled data, and h_1 and h_2 denote the base learners in the ensemble. Each dot denotes one instance. The dots in the left graph represent the instances in the original labeled data. The red and blue dots denote the positive and negative instances, respectively. The yellow dots in the right graph denote the instances in the transformed data, which are unlabeled. The colored circle around the yellow dots denote their labels in the original data. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$l_{2,1}$ -norm of $\mathbf{W}$ are denoted by $\text{Tr}[\mathbf{W}]$, $\mathbf{W}^T$, $\mathbf{W}^{-1}$, and $\|\mathbf{W}\|_{2,1}$, respectively. Note that vectors and matrices are represented in bold font.

3.2. Data transformation

Considering that class labels do not directly related to the prediction power of ensemble, we transform the original labeled data to correct prediction space in this subsection.

The data transformation is expressed as follows.

Let $\tilde{h}_j(\mathbf{x}_i)$ denote the probability of the i th instance, $\mathbf{x}_i$, belonging to its true class label y_i , which is predicted by the base learner h_j . In other words, $\tilde{h}_j(\mathbf{x}_i)$ represents the probability of correct prediction by base learner h_j for $\mathbf{x}_i$. For example, the prediction vector of a base learner h for $\mathbf{x}$ is $(0.7/\text{positive class}, 0.3/\text{negative class})$. If $\mathbf{x}$'s true class label is positive, then the probability of correct prediction by h for $\mathbf{x}$ is 0.7 (i.e., $\tilde{h}(\mathbf{x}) = 0.7$).

The dataset $\mathbf{S}$ can be transformed into $\mathbf{X}'$ using ensemble H , which is expressed as follows:

$$\mathbf{X}' = [\mathbf{x}'_1, \mathbf{x}'_2, \dots, \mathbf{x}'_n] = \begin{bmatrix} \tilde{h}_1(\mathbf{x}_1) & \dots & \tilde{h}_1(\mathbf{x}_n) \\ \vdots & \ddots & \vdots \\ \tilde{h}_k(\mathbf{x}_1) & \dots & \tilde{h}_k(\mathbf{x}_n) \end{bmatrix}. \quad (6)$$

Note that the transformed dataset $\mathbf{X}' \in \mathbb{R}^{k \times n}$ leverages the distribution of correct prediction by ensemble H for $\mathbf{S}$. Each column in $\mathbf{X}'$, i.e., $\mathbf{x}'_i = [\tilde{h}_1(\mathbf{x}_i), \dots, \tilde{h}_k(\mathbf{x}_i)]^T \in \mathbb{R}^k$, represents $\mathbf{x}_i$'s distribution of correct prediction by the ensemble. Additionally, each row in $\mathbf{X}'$, i.e., $\tilde{h}_j(\mathbf{X}) = [\tilde{h}_j(\mathbf{x}_1), \dots, \tilde{h}_j(\mathbf{x}_n)]$, describes the distribution of correct prediction by base learner h_j .

This transformation is illustrated as shown in Fig. 1. In right graph of Fig. 1, each dimension describes the correct prediction power of the corresponding base learner. All dimensions describe the whole prediction power of ensemble. Furtherly, Fig. 1 demonstrates that in the transformed data $\mathbf{X}'$, dots that share similar prediction power may be clustered together. This cluster structure can be seen as a kind of intrinsic structure globally describing the correct-prediction power of ensemble. Keeping this global prediction distribution i.e., the cluster structure, invariant, the dimension of ensemble can be reduced with less or even no loss of ensemble's prediction performance. As the transformed data $\mathbf{X}'$ is unlabeled, the ensemble selection problem on $\mathbf{S}$ becomes an unsupervised ensemble selection on $\mathbf{X}'$.

Moreover, it is observed from Fig. 1 that the dots with differently colored circles are clustered together in transformed data $\mathbf{X}'$. This shows that the instances with similar prediction distributions

may cluster together regardless of whether they have the same class labels. Last but not least, we find that in the transformed data $\mathbf{X}'$, small perturbation of each dot will not destroy the global cluster structure, then it will not disturb ensemble selection. Thus, the transformed data is less sensitive to the local prediction result of each base learner.

However, there exists an apparent limitation of our proposed transformation. Specifically, the prediction of the base learner should be a probability distribution; otherwise it is infeasible to transform original data into unlabeled prediction space.

3.3. Proposed objective function

After ensemble transformation, the global prediction power of the ensemble is described in the unlabeled transformed data. In the transformed data, the cluster structure of the instances leverages the global prediction power of the ensemble. Therefore, maintaining the cluster structure as invariant is equivalent to maintaining the prediction distribution as invariant. Thus we design our objective function based on the idea of reducing the redundant base learners without changing the cluster structure.

To learn the prediction distribution of the ensemble and to reduce its dimensionality, we utilize nonnegative matrix factorization with orthogonality constraints to learn the cluster structure of the transformed data. Spectral clustering techniques have been demonstrated to be effective in detecting the cluster structure of data and have recently received significant research attention [18]. As a result, we use spectral clustering to learn the pseudo cluster-labels in the transformed data, which guide the process of inferring the base learner selection matrix. To select the most competent base learners, structural sparsity is adopted to perform dimensionality reduction on the transformed data. The objective function of our proposed ES-JSS is described as follows:

$$\min_{\mathbf{W}, \mathbf{E}^T \mathbf{E} = \mathbf{I}, \mathbf{E} \geq 0} \text{Tr}[\mathbf{E}^T \mathbf{L} \mathbf{E}] + \lambda \|\mathbf{W}^T \mathbf{X}' - \mathbf{E}^T\|_F^2 + \theta \|\mathbf{W}\|_{2,1}, \quad (7)$$

where the weight matrix $\mathbf{W} \in \mathbb{R}^{k \times c}$, the pseudo cluster-label matrix $\mathbf{E} \in \mathbb{R}^{n \times c}$, $\lambda \in \mathbb{R}_+$, and $\theta \in \mathbb{R}_+$ are parameters.

In the spectral clustering formulation $\text{Tr}[\mathbf{E}^T \mathbf{L} \mathbf{E}]$, pseudo cluster label matrix $\mathbf{E}$ is learned on the transformed data $\mathbf{X}'$, i.e., spectral clustering relies on the similarity matrix computed from the transformed data $\mathbf{X}'$. The transformed data with the cluster label matrix contain information on not only correct prediction but also on the competence of the base learners.

In the regression-based formulation $\|\mathbf{W}^T \mathbf{X}' - \mathbf{E}^T\|_F^2$, each row of $\mathbf{X}'$, i.e., $\tilde{h}_j(\mathbf{X}) = [\tilde{h}_j(\mathbf{x}_1), \dots, \tilde{h}_j(\mathbf{x}_n)]$, can be perceived as a feature

that describes the competence of the corresponding base learner. Clustering is performed by jointly minimizing the first two terms which shows that the pseudo cluster labels $\mathbf{E}$ are predicted by the transformed data along with the data in the low-dimensional subspace. These first two terms ensure that the base learners share a common clustering structure, i.e., prediction distribution in a low-dimensional space.

To guarantee the property of row-wise sparseness, we adopt the $l_{2,1}$ -norm as a regularization term for the objective function. Robust ensemble selection is performed by jointly minimizing the last two terms. Specifically, the joint minimization of the regression term and $l_{2,1}$ -norm regularization term enables $\mathbf{W}$ to evaluate the correlation between pseudo cluster labels and base learners, particularly suited to ensemble selection. Note that the row $\mathbf{w}^i$ in $\mathbf{W}$ shrinks to zeros if the i th base learner is less correlated with the pseudo cluster-label matrix $\mathbf{E}$. Thus, we can filter out the base learners whose rows of $\mathbf{W}$ are approximately to zero-rows.

4. Optimization

The optimization problem in Eq. (5), which involves the $l_{2,1}$ -norm is non-smooth and difficult to solve directly. Consequently, we propose an iterative optimization algorithm.

4.1. Efficient iterative algorithm

Update $\mathbf{W}$: we temporarily remove the terms that are irrelevant to $\mathbf{W}$ in Eq. (5). Then, we obtain

$$\min_{\mathbf{W}} \lambda \|\mathbf{W}^T \mathbf{X}' - \mathbf{E}^T\|_F^2 + \theta \|\mathbf{W}\|_{2,1}. \quad (8)$$

To obtain the optimization value in Eq. (6), we adopt the gradient descent method, which we express as follows:

$$\begin{aligned} & \lambda \mathbf{X}'(\mathbf{X}'^T \mathbf{W} - \mathbf{E}) + \theta \mathbf{D} \mathbf{W} = 0 \\ \Leftrightarrow & \lambda \mathbf{X}' \mathbf{X}'^T \mathbf{W} - \lambda \mathbf{X}' \mathbf{E} + \theta \mathbf{D} \mathbf{W} = 0 \\ \Leftrightarrow & \mathbf{W} = \lambda (\lambda \mathbf{X}' \mathbf{X}'^T + \theta \mathbf{D})^{-1} \mathbf{X}' \mathbf{E}, \end{aligned} \quad (9)$$

where $\mathbf{D} \in \mathbb{R}^{k \times k}$ represents a diagonal matrix with elements $D_{ii} = \frac{1}{2\|\mathbf{w}^i\|_2}$.

Update $\mathbf{E}$: using nonnegative and orthogonality constraints, we reformulate the objective function in Eq. (5) as follows:

$$\min_{\mathbf{W}, \mathbf{E}} \text{Tr}[\mathbf{E}^T \mathbf{L} \mathbf{E}] + \lambda \|\mathbf{W}^T \mathbf{X}' - \mathbf{E}^T\|_F^2 + \theta \|\mathbf{W}\|_{2,1} + \frac{\gamma}{2} \|\mathbf{E}^T \mathbf{E} - \mathbf{I}_c\|_F^2, \quad (10)$$

s.t. $\mathbf{E} \geq 0$,

where $\gamma > 0$ represents a parameter for controlling the orthogonality condition. In practice, γ should be large enough to ensure that orthogonality is satisfied.

By substituting $\mathbf{W}$ into Eq. (7) and removing the terms that are irrelevant to $\mathbf{E}$, the problem in Eq. (8) can be rewritten as follows:

$$\min_{\mathbf{E} \geq 0} \text{Tr}[\mathbf{E}^T \mathbf{M} \mathbf{E}] + \frac{\gamma}{2} \|\mathbf{E}^T \mathbf{E} - \mathbf{I}_c\|_F^2, \quad (11)$$

where $\mathbf{M} = \mathbf{L} + \lambda[\mathbf{I}_n - \lambda \mathbf{X}'(\mathbf{X}' \mathbf{L} \mathbf{X}'^T + \lambda \mathbf{X}' \mathbf{X}'^T + \theta \mathbf{D})^{-1} \mathbf{X}']$.

A multiplicative updating rule [17,29] can be used to solve the problem in Eq. (9). We obtain the optimization $\mathbf{E}$ as follows:

$$E_{ij} = E_{ij} \frac{(\gamma \mathbf{E})_{ij}}{(\mathbf{M} \mathbf{E} + \gamma \mathbf{E} \mathbf{E}^T \mathbf{E})_{ij}}. \quad (12)$$

Next, we normalize $\mathbf{E}$ such that $(\mathbf{E}^T \mathbf{E})_{ii} = 1$ and $i = 1, \dots, c$.

The unsupervised ensemble selection technique is designed using $\mathbf{X}'$, i.e., the pseudo cluster-labels are learned and the base learners selected simultaneously in the procedure of ensemble selection on $\mathbf{X}'$ by optimizing Eq. (5). Figure 2 presents a schematic illustration of the proposed approach.

Based on the above analysis, the ensemble selection algorithm can be given by shown in Algorithm 1.

Algorithm 1: Algorithm (ES-JSS).

Input: Set of trained base learners $H = \{h_i(x)\}_{i=1}^k$; Validation set $\mathbf{S} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^n$; Latent cluster number c ; Top p base learners; regularization parameter λ, θ ; a large enough γ .

Output: p base learners for the dataset.

- 1 Transform $\mathbf{S}$ to $\mathbf{X}'$ by H ;
- 2 Calculate $\mathbf{B}$ on transformed data $\mathbf{X}'$ by Eq. (4), and then calculate $\mathbf{L} = \mathbf{A} - \mathbf{B}$;
- 3 Iteration step $t = 0$; Initialize $\mathbf{E}_t \in \mathbb{R}^{n \times c}$ and set $\mathbf{D}_t \in \mathbb{R}^{k \times k}$ as an identity matrix;
- 4 **repeat**
- 5 Update $\mathbf{E}_{t+1}$ by Eq. (10);
- 6 Update $\mathbf{W}_{t+1}$ by Eq. (7);
- 7 Update the diagonal matrix $\mathbf{D}$ with

$$\mathbf{D}_{t+1} = \begin{bmatrix} \frac{1}{2\|\mathbf{w}_{t+1}^1\|_2} & & \\ & \dots & \\ & & \frac{1}{2\|\mathbf{w}_{t+1}^k\|_2} \end{bmatrix}$$
- 8 $t = t + 1$;
- 9 **until** convergence;
- 10 Sort all k base learners in descending order according to $\|\mathbf{w}_t^i\|_2$ ($i = 1, \dots, k$) and select the top- p ranked base learners.

4.2. Convergence analysis

The proposed iterative procedure shown in Algorithm 1 can be solved to converge with the optimal solutions through the following analysis.

Theorem 1: The alternative updating rules in the gradient descent algorithm monotonically decrease Eq. (5) in each iteration.

Proof:

For convenience, we rewrite Eq. (5) as follows:

$$\Psi(\mathbf{W}, \mathbf{E}) = \text{Tr}[\mathbf{E}^T \mathbf{L} \mathbf{E}] + \lambda \|\mathbf{W}^T \mathbf{X}' - \mathbf{E}^T\|_F^2 + \theta \|\mathbf{W}\|_{2,1}. \quad (13)$$

Firstly, it is relatively straightforward to establish that Eq. (11) monotonically decreases only with respect to $\mathbf{E}$ by using the auxiliary function method [29]. Therefore, we obtain the following:

$$\Psi(\mathbf{E}_{t+1}, \mathbf{W}_t) \leq \Psi(\mathbf{E}_t, \mathbf{W}_t). \quad (14)$$

We define $\aleph(\mathbf{W}) = \theta \text{Tr}[\mathbf{W}^T \mathbf{D} \mathbf{W}]$. Then Eq. (5) can be rewritten only with respect to $\mathbf{W}$ as follows:

$$\min_{\mathbf{W}} \lambda \|\mathbf{W}^T \mathbf{X}' - \mathbf{E}^T\|_F^2 + \aleph(\mathbf{W}). \quad (15)$$

By defining $\mathbf{W}_{t+1}$ in terms of the t th iteration with $\mathbf{E}_t$ fixed, we obtain the following:

$$\begin{aligned} \mathbf{W}_{t+1} &= \arg \min_{\mathbf{W}} \lambda \|\mathbf{W}^T \mathbf{X}' - \mathbf{E}_t^T\|_F^2 + \aleph(\mathbf{W}) \\ \Rightarrow \mathbf{W}_{t+1} &= \arg \min_{\mathbf{W}} \lambda \|\mathbf{W}_t^T \mathbf{X}' - \mathbf{E}_t^T\|_F^2 + \theta \text{Tr}[\mathbf{W}_t^T \mathbf{D}_t \mathbf{W}_t]. \end{aligned} \quad (16)$$

Therefore,

$$\begin{aligned} & \lambda \|\mathbf{W}_{t+1}^T \mathbf{X}' - \mathbf{E}_t^T\|_F^2 + \theta \text{Tr}[\mathbf{W}_{t+1}^T \mathbf{D}_t \mathbf{W}_{t+1}] \\ & \leq \lambda \|\mathbf{W}_t^T \mathbf{X}' - \mathbf{E}_t^T\|_F^2 + \theta \text{Tr}[\mathbf{W}_t^T \mathbf{D}_t \mathbf{W}_t]. \end{aligned} \quad (17)$$

It is also straight to obtain

$$\begin{aligned} & \lambda \|\mathbf{W}_{t+1}^T \mathbf{X}' - \mathbf{E}_t^T\|_F^2 + \theta \sum_i \frac{\|\mathbf{w}_{t+1}^i\|_2^2}{2\|\mathbf{w}_t^i\|_2} \leq \lambda \|\mathbf{W}_t^T \mathbf{X}' - \mathbf{E}_t^T\|_F^2 + \theta \sum_i \frac{\|\mathbf{w}_t^i\|_2^2}{2\|\mathbf{w}_t^i\|_2} \\ \Rightarrow & \lambda \|\mathbf{W}_{t+1}^T \mathbf{X}' - \mathbf{E}_t^T\|_F^2 + \theta \|\mathbf{W}_{t+1}\|_{2,1} - \theta \left(\|\mathbf{W}_{t+1}\|_{2,1} - \sum_i \frac{\|\mathbf{w}_{t+1}^i\|_2^2}{2\|\mathbf{w}_t^i\|_2} \right) \\ & \leq \lambda \|\mathbf{W}_t^T \mathbf{X}' - \mathbf{E}_t^T\|_F^2 + \theta \|\mathbf{W}_t\|_{2,1} - \theta \left(\|\mathbf{W}_t\|_{2,1} - \sum_i \frac{\|\mathbf{w}_t^i\|_2^2}{2\|\mathbf{w}_t^i\|_2} \right). \end{aligned} \quad (18)$$

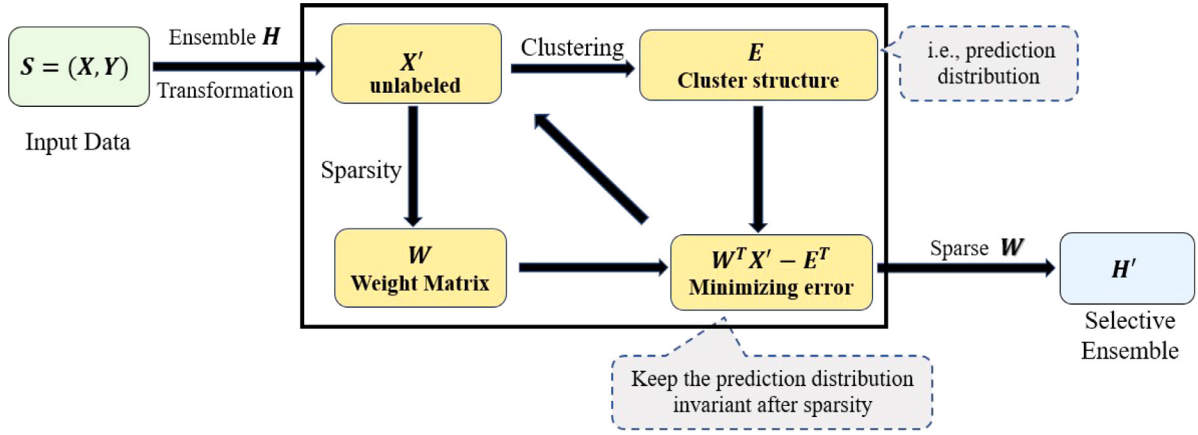


Fig. 2. Schematic illustration of the proposed approach.

According to Lemma 1 in [30], i.e., for any nonzero vectors, $\mathbf{u}, \mathbf{u}_t$, $\|\mathbf{u}\|_2 - \frac{\|\mathbf{u}\|_2^2}{2\|\mathbf{u}_t\|_2} \leq \|\mathbf{u}_t\|_2 - \frac{\|\mathbf{u}_t\|_2^2}{2\|\mathbf{u}_t\|_2}$, we obtain

$$\|\mathbf{W}_{t+1}\|_{2,1} - \sum_i \frac{\|\mathbf{w}_t^i\|_2^2}{2\|\mathbf{w}_t^i\|_2} \leq \|\mathbf{W}_t\|_{2,1} - \sum_i \frac{\|\mathbf{w}_t^i\|_2^2}{2\|\mathbf{w}_t^i\|_2}. \quad (19)$$

Thus, we obtain the following:

$$\lambda \|\mathbf{W}_{t+1}^T \mathbf{X}' - \mathbf{E}_t^T\|_F^2 + \theta \|\mathbf{W}_{t+1}\|_{2,1} \leq \lambda \|\mathbf{W}_t^T \mathbf{X}' - \mathbf{E}_t^T\|_F^2 + \theta \|\mathbf{W}_t\|_{2,1}. \quad (20)$$

Therefore,

$$\Psi(\mathbf{E}_t, \mathbf{W}_{t+1}) \leq \Psi(\mathbf{E}_t, \mathbf{W}_t). \quad (21)$$

Based on above analysis and Eqs. (12), and (18), we arrive at:

$$\Psi(\mathbf{E}_{t+1}, \mathbf{W}_{t+1}) \leq \Psi(\mathbf{E}_t, \mathbf{W}_t). \quad (22)$$

Therefore, $\Psi(\mathbf{W}, \mathbf{E})$ monotonically decreases using the updating rules in Algorithm 1, and Theorem 1 is proved.

Using Theorem 1, we can establish that the iterative approach in Algorithm 1 makes the objective function converge to its global optimal solutions. As a result, the proposed optimization algorithm is efficient.

4.3. Computational complexity analysis

Here we briefly analyze the computational complexity of ES-JSS in Algorithm 1. The computational cost of constructing $\mathbf{X}'$ using H for $\mathbf{X}$ is $O(nk)$. In each iteration, the cost of updating $\mathbf{E}_{t+1}$ with $E_{ij} = E_{ij} \frac{(\gamma \mathbf{E})_{ij}}{(\mathbf{M}\mathbf{E} + \gamma \mathbf{E}\mathbf{E}^T \mathbf{E})_{ij}}$ must be $O(k^3 + nk^2 + kn^2)$, and the cost of updating $\mathbf{W}_{t+1}$ with $\mathbf{W} = \lambda(\lambda \mathbf{X}' \mathbf{X}'^T + \theta \mathbf{D})^{-1} \mathbf{X}' \mathbf{E}$ is $O(k^3 + nk^2 + kn^2 + knc + kc^2)$. Suppose that $c \ll n$ and $c \ll k$; then, the overall cost of the proposed ES-JSS is approximately $O(t(k^3 + kn^2))$, where t represents the number of iterations.

5. Experiments

In this section, we report on extensive experiments to evaluate the performance of the proposed ES-JSS. First, we introduce experimental setting including the datasets used. Second, we present a performance comparison with a number of state-of-the-art algorithms. Finally, we illustrate the parameter sensitivity and convergence of ES-JSS.

5.1. Experimental setting

Baseline: To validate the effectiveness of ES-JSS, we compared it with six diverse dynamic ensemble selection methods: two meth-

ods focusing on region of competence, KNORA-E [20], and Fire-DES++ [21], and four methods focusing on estimation of competence, META-DES [23], DESP [22], KNOP [12], and DES-MI [13]. We also compared ES-JSS with six diverse static ensemble selection methods: four order-based methods, MD [7], CP [7], Kappa [6], and RE [6], and two optimization-based methods, PEP [9], and DOME [10]. Further, we compared ES-JSS with four known ensemble methods, Bagging [2], GBDT [4], RF [3], and XGBoost [31]. Of the 16 compared algorithms, some are classic and some were proposed recently with state-of-the-art performance. Therefore, these diverse algorithms can effectively highlight the performance of the proposed ES-JSS from different perspective.

All the codes for the dynamic ensemble selection techniques, such as Meta-DES, KNORA-E, DESP, KNOP and DES-MI were from the DESlib library [33]. The hyperparameters of such dynamic techniques were set as their default values reported in [33] and their corresponding original works. Specifically, the decisions of the selected base classifiers in KNORA-E and KNOP are aggregated using the majority voting rule and the size of the region of competence was equally set to 7 (as suggested in [11]). In META-DES, five sets of meta-features for the estimation of the classifier competence are used, as reported in [23]. The hyperparameters of the static techniques were also assigned their corresponding suggested values in their original works, except the number of selected base learners. In this study, 20 base learners were selected for all static ensemble selections, except for PEP. PEP's selective size cannot be preset because it is a bi-objective optimization method. For known ensemble methods, such as GBDT and Bagging, 100 iterations are used (i.e., 100 base classifier are generated). For RF, the number of trees was set to 100 and the number of leaves was set to the square root of the number of features.

Datasets: Our experiments were performed on 10 datasets from UCI¹ and 15 datasets from KEEL², as described in Table 1. These datasets all noticeably differ in their instance sizes, feature numbers and label numbers, thus, they serve as a good test bed for a comprehensive evaluation.

The datasets were randomly divided as follows: 50% of data for ensemble generation, 25% for dynamic or static selection, and 25% for test set, (i.e., adhering to the settings presented by Cruz et al. [11], except for Fire-DES++ and META-DES). To ensure the performance of Fire-DES++, we used 75% of the data for ensemble generation (half of the 75% for dynamic selection) and 25% for test data. For the proposed META-DES, 50% of the training data was used in

¹ <http://archive.ics.uci.edu/ml>

² <https://sci2s.ugr.es/keel/datasets.php>

Table 1
Description of the selected datasets.

Datasets	Referred to as	Instances	Features	Classes	Source
adult	adult	45,222	14	2	KEEL
australian	australian	690	14	2	KEEL
cancer-683	cancer	683	9	2	UCI
census	census	142,521	41	3	KEEL
coil2000	coil2000	9,822	85	2	KEEL
column_3c_weka	column	310	6	3	UCI
credit	credit	30,000	23	2	UCI
crowd	crowd	10,845	28	2	UCI
eeg	eeg	14,980	14	2	UCI
fars	fars	100,968	29	8	KEEL
flare	flare	1,066	11	6	KEEL
FPS-5	FPS-5	2,688	3,208	6	UCI
german	german	1,000	20	2	KEEL
letter	letter	20,000	16	26	UCI
magic	magic	19,020	10	2	KEEL
marketing	market	6,876	13	9	KEEL
nursery	nursery	12,690	8	5	KEEL
optdigits	optdigits	5,620	64	10	UCI
ring	ring	7,400	20	2	KEEL
sick	sick	2,800	26	2	UCI
spambase	spambase	4,597	57	2	KEEL
thyroid	thyroid	7,200	21	3	KEEL
twonorm	twonorm	7,400	20	2	KEEL
waveform	waveform	5,000	21	3	UCI
winequality-red	wine	1,599	11	11	KEEL

the meta-training process and 50% for the generation of the pool of classifiers. Then an ensemble is trained on training set, an ensemble selection method is conducted on the validation set, and finally tested on the test set. For all evaluations, we report the average results with their standard deviation for five runs per dataset.

Parameters: The experiments were performed using Python 3.5 with the scikit-learn library to train the base learners. Similar to the previous ensemble selection works [9], the C4.5 Decision Tree [32] was chosen as the base learner to form an ensemble $H = \{h_i(x)\}_{i=1}^k$, where $k = 100$. The use of Decision Tree as base classifier comes from the following observations. 1) The bagging ensemble built by weak and unstable base learners, such as Decision trees, Perceptrons and Bayesian Networks, can achieve better performance, just as reported in [2,23]. 2) Previous works on ensemble selection usually adopt weak base learners, such as Decision trees or Perceptrons, as base classifier [8,9,21,23]. 3) Our proposed ES-JSS is robust to the type of base learner, just as experimentally demonstrated in Section 5.3. Thus, in this study we use DT as base learners similar with previous works [8,9]. Similar results can be obtained based on other weak base learners, such as Perceptron and Bayesian.

We coded ES-JSS using the PyTorch system and report the average results on each dataset using fixed parameters. The number of iterations used for ensemble selection were set to 100. With reference to Algorithm 1, ES-JSS has five parameters: λ , θ , p , γ , and c . γ is automatically set by the system to guarantee orthogonality. In our parameter sensitivity study, presented in Appendix, ES-JSS was less sensitive to the values of λ , θ , and c . Therefore, we fixed $\lambda = 0.05$, $\theta = 2.0$, and $c = 2$ for each dataset. On the contrary, ES-JSS was sensitive to p , just as claimed in Subsection 5.4. Therefore, we set $p = \{1, 2, 3, \dots, 99, 100\}$, and a grid search strategy was used to determine the optimal p . Usually, $p \in [15, 50]$ is as the suggested interval.

5.2. Performance Evaluation

In this subsection, we compare the prediction performances of the ensemble selection methods. We present the results in Tables 2, 3, 4 and 5, which show the comparisons with the eight state-of-the-art dynamic selection methods, six static ensemble selection methods, four known ensemble methods, and two standard

unsupervised feature selection methods, respectively. In these tables, the best results on each dataset are highlighted in bold font. Additionally, we list the count of best performance of each method, indicated by the row 'count of bests'. Furthermore, we used the Sign test [34] to conduct a pairwise comparison between algorithms over all datasets. The Sign-Test works upon the number of wins, ties and losses obtained by an algorithm over the baseline, indicated by the row of 'Wins of ES-JSS'. The algorithm is deemed statistically better if its number of wins plus half of the number of ties is higher than a critical value. Through t-test for pairwise comparison on each data set, we indicate the performance of ES-JSS against other baselines on each dataset.

Table 2 presents the detailed results for ES-JSS, compared with those of dynamic ensemble selection methods. ES-JSS achieved the best performance on 64% (16/25) of the datasets, whereas none of the other methods reached more than 20% (5/25), as indicated by the 'Number the bests' row. In the 'Wins of ES-JSS' evaluation, ES-JSS outperformed KNORA-E on 84%(21/25) of the datasets and outperformed the others by at least 60%(15/25) at least; e.g., META-DES(15/25) and KNOP(15/25). From the t-tests with a significance level of 0.05, ES-JSS was not significantly inferior to the compared methods on 80% of the datasets. These results show that ES-JSS outperformed the dynamic ensemble selection algorithms in most cases.

The observations from Tables 3 and 4 are similar. Therefore, for brevity, we do not discuss them here. From the results of Tables 2-4, we can make the following observations:

- ES-JSS is more robust than dynamic ensemble selection methods in most cases. This shows that it is robust and efficient for conducting ensemble selection on global prediction spaces (i.e., transformed data). Specifically, ensemble selection based on global prediction distribution can reduce the sensitivity of selective ensemble and make our ES-JSS algorithm more robust and efficient.
- As an unsupervised static ensemble selection algorithm, ES-JSS is superior to supervised static ensemble selection methods, such as MD, CP, Kappa, and RE, in most cases. This shows that the class label is not the crucial factor in static ensemble selection; rather, the prediction power of each base learner is important.

Table 2

Mean and standard deviation results for the accuracy of the proposed ES-JSS and six diverse dynamic ensemble selection methods. For each dataset, the best values are in bold font. '•' and '◦' respectively denote that ES-JSS was significantly better or worse than the given method using a t-test with a confidence level of 0.05. The 'Number of bests' row lists the numbers of datasets for which the best accuracy was obtained. The 'Wins of ES-JSS' row lists the numbers of datasets for which ES-JSS was significantly better than the corresponding method using the t-test (a tie was counted as 0.5 wins).

Datasets	ES-JSS	META-DES	KNORA-E	DESP	KNOP	DES-MI	FIRE+
adult	84.37 ± 0.10	83.96 ± 0.78	82.77 ± 0.15 •	84.33 ± 0.21	84.37 ± 0.26	84.11 ± 0.25	77.56 ± 10.94 •
australian	86.61 ± 1.33	86.36 ± 2.23	85.61 ± 2.03	86.65 ± 2.12	86.42 ± 2.13	86.18 ± 2.01	85.66 ± 1.78
cancer	98.46 ± 0.64	95.85 ± 1.01 •	95.32 ± 1.35 •	95.5 ± 1.18 •	95.73 ± 1.00 •	96.08 ± 1.10 •	96.32 ± 1.14 •
census	95.70 ± 0.02	95.61 ± 0.08 •	95.09 ± 0.09 •	95.62 ± 0.07 •	95.64 ± 0.09 •	95.57 ± 0.08 •	95.13 ± 0.08 •
coil2000	94.10 ± 0.10	92.79 ± 0.55 •	91.80 ± 0.40 •	92.41 ± 0.43 •	92.75 ± 0.32 •	92.05 ± 0.51 •	66.47 ± 7.67 •
cloumn	86.62 ± 2.38	81.67 ± 3.47 •	80.00 ± 4.20 •	82.56 ± 3.09 •	82.82 ± 2.91 •	83.85 ± 3.88 •	82.95 ± 4.52 •
credit	81.24 ± 0.13	80.65 ± 0.32 •	77.26 ± 0.38 •	80.74 ± 0.27 •	80.79 ± 0.25 •	80.49 ± 0.30 •	45.85 ± 15.50 •
crowd	91.94 ± 0.14	92.70 ± 0.38 ◦	91.68 ± 0.30 •	92.29 ± 0.32 ◦	92.40 ± 0.24 ◦	92.52 ± 0.32 ◦	92.87 ± 0.53 ◦
eeg	90.04 ± 0.33	91.33 ± 0.57 ◦	89.78 ± 0.71	91.05 ± 0.69 ◦	90.85 ± 0.47 ◦	90.08 ± 0.53	91.12 ± 0.30 ◦
fars	76.23 ± 0.06	75.84 ± 0.31 •	75.21 ± 0.28 •	75.91 ± 0.19 •	76.08 ± 0.25 •	75.88 ± 0.19 •	75.74 ± 0.61 •
flare	52.25 ± 0.81	48.11 ± 2.37 •	46.89 ± 3.20 •	47.52 ± 2.01 •	49.17 ± 1.56 •	48.31 ± 2.48 •	49.04 ± 2.82 •
FPS-5	79.09 ± 0.42	79.61 ± 1.22	75.06 ± 1.85 •	78.99 ± 1.51	79.36 ± 1.50	78.86 ± 1.21	81.08 ± 0.84 ◦
german	77.42 ± 1.62	75.24 ± 3.15 •	72.04 ± 1.76 •	75.36 ± 2.26 •	75.48 ± 2.30 •	74.40 ± 3.06 •	74.52 ± 3.09 •
letter	91.55 ± 0.18	93.99 ± 0.48 ◦	90.58 ± 0.65 •	93.32 ± 0.58 ◦	93.58 ± 0.54 ◦	93.65 ± 0.63 ◦	94.21 ± 0.39 ◦
magic	88.03 ± 0.06	87.26 ± 0.42 •	85.38 ± 0.49 •	87.28 ± 0.49 •	87.26 ± 0.50 •	86.98 ± 0.49 •	86.61 ± 0.52 •
market	31.53 ± 0.42	30.34 ± 1.25 •	28.79 ± 1.22 •	30.81 ± 1.05 •	30.76 ± 1.33	30.56 ± 1.13	29.59 ± 1.60 •
nursery	99.12 ± 0.10	99.61 ± 0.15 ◦	99.59 ± 0.12 ◦	99.44 ± 0.16 ◦	99.51 ± 0.16 ◦	99.43 ± 0.17 ◦	99.73 ± 0.13 ◦
optdigits	95.36 ± 0.21	96.23 ± 0.59 ◦	95.56 ± 0.60	95.96 ± 0.63 ◦	96.03 ± 0.63 ◦	96.20 ± 0.63 ◦	96.54 ± 0.98 ◦
ring	94.70 ± 0.19	94.92 ± 0.99	94.08 ± 0.86 •	94.16 ± 1.15	94.65 ± 1.08	93.85 ± 0.95 •	93.83 ± 0.34 •
sick	98.31 ± 0.25	96.93 ± 3.58	97.77 ± 0.60	97.84 ± 0.64 •	97.97 ± 0.60	97.93 ± 0.69	98.13 ± 0.78
spambase	94.34 ± 0.15	94.48 ± 1.00	93.87 ± 0.90	94.07 ± 1.11	94.16 ± 0.97	94.21 ± 1.00	94.14 ± 0.60
thyroid	99.77 ± 0.02	99.57 ± 0.10 •	99.58 ± 0.08 •	99.56 ± 0.10 •	99.60 ± 0.08 •	99.58 ± 0.09 •	99.63 ± 0.12 •
twonorm	96.71 ± 0.22	96.60 ± 0.42	95.25 ± 0.48 •	96.60 ± 0.43	96.66 ± 0.46	96.14 ± 0.64 •	95.50 ± 0.45 •
waveform	83.78 ± 0.45	83.62 ± 0.86	80.02 ± 0.96 •	83.65 ± 0.75	83.73 ± 0.89	83.12 ± 1.03 •	83.28 ± 1.13
wine	71.72 ± 0.97	71.24 ± 2.34	65.34 ± 2.74 •	69.12 ± 2.57 •	69.65 ± 2.73 •	69.74 ± 2.49 •	70.99 ± 1.71
Number of bests	16	3	0	1	1	0	5
Wins of ES-JSS		15	21	16	15	17.5	15.5

Table 3

Mean and standard deviation results for the accuracy of ES-JSS and six diverse static ensemble selection methods. For each dataset, the best values are in bold font. '•' and '◦' respectively denote that ES-JSS was significantly better or worse than the given method using a t-test with a confidence level of 0.05. The 'Number of bests' row lists the numbers of datasets for which the best accuracy was obtained. The 'Wins of ES-JSS' row lists the numbers of datasets for which ES-JSS was significantly better than the corresponding method using the t-test (a tie was counted as 0.5 wins).

Datasets	ES-JSS	MD	CP	Kappa	RE	DOMEP	PEP
adult	84.37 ± 0.10	80.19 ± 0.09 •	80.17 ± 0.10 •	80.22 ± 0.11 •	80.26 ± 0.11 •	84.26 ± 0.04 •	84.30 ± 0.33
australian	86.61 ± 1.33	86.25 ± 0.25	86.17 ± 0.75	86.48 ± 0.56	86.46 ± 0.51	87.54 ± 0.95 ◦	89.60 ± 1.73 ◦
cancer	98.46 ± 0.64	96.51 ± 0.35 •	96.59 ± 0.43 •	96.34 ± 0.69 •	96.28 ± 0.42 •	97.22 ± 1.58 •	99.19 ± 0.31 ◦
census	95.70 ± 0.02	84.35 ± 0.09 •	84.36 ± 0.12 •	84.38 ± 0.11 •	84.37 ± 0.11 •	95.53 ± 0.07 •	95.57 ± 0.06 •
coil2000	94.10 ± 0.10	65.48 ± 0.83 •	65.65 ± 0.41 •	65.87 ± 0.42 •	65.58 ± 0.71 •	93.31 ± 0.04 •	92.36 ± 0.47 •
cloumn	86.62 ± 2.38	82.88 ± 1.32 •	82.25 ± 1.15 •	81.87 ± 1.24 •	82.10 ± 1.60 •	87.10 ± 0.00	83.25 ± 3.85 •
credit	81.24 ± 0.13	71.06 ± 0.20 •	71.02 ± 0.15 •	71.04 ± 0.16 •	71.07 ± 0.25 •	80.01 ± 0.47 •	80.20 ± 0.25 •
crowd	91.94 ± 0.14	78.86 ± 0.66 •	78.29 ± 0.96 •	78.48 ± 0.99 •	78.26 ± 0.76 •	91.74 ± 0.40	91.82 ± 0.55
eeg	90.04 ± 0.33	90.45 ± 0.15 ◦	90.54 ± 0.25 ◦	90.59 ± 0.14 ◦	90.48 ± 0.21 ◦	90.36 ± 0.24 ◦	90.59 ± 0.68 ◦
fars	76.23 ± 0.06	39.03 ± 1.25 •	39.53 ± 2.64 •	40.59 ± 1.55 •	40.28 ± 1.94 •	75.89 ± 0.27 •	75.90 ± 0.23 •
flare	52.25 ± 0.81	44.86 ± 0.86 •	45.04 ± 1.39 •	45.21 ± 0.36 •	45.32 ± 0.80 •	48.66 ± 2.96 •	52.37 ± 0.70
FPS-5	79.09 ± 0.42	79.50 ± 0.36 ◦	79.46 ± 0.63	79.73 ± 0.31 ◦	79.89 ± 0.50 ◦	78.31 ± 0.79 •	78.62 ± 1.46
german	77.42 ± 1.62	68.56 ± 0.85 •	68.31 ± 1.16 •	68.82 ± 0.88 •	69.01 ± 0.82 •	75.48 ± 1.17 •	74.25 ± 1.25 •
letter	91.55 ± 0.18	92.75 ± 0.20 ◦	92.73 ± 0.16 •	92.66 ± 0.23 •	92.69 ± 0.15 •	92.64 ± 0.52 •	92.46 ± 0.63 •
magic	88.03 ± 0.06	85.93 ± 0.14 •	85.99 ± 0.09 •	86.01 ± 0.13 •	86.01 ± 0.11 •	86.95 ± 0.43 •	87.17 ± 0.45 •
market	31.53 ± 0.42	28.39 ± 0.28 •	28.46 ± 0.37 •	28.33 ± 0.25 •	28.48 ± 0.32 •	31.24 ± 0.35	34.03 ± 0.35 ◦
nursery	99.12 ± 0.10	92.28 ± 0.33 •	92.21 ± 0.28 •	92.26 ± 0.24 •	92.35 ± 0.38 •	99.33 ± 0.23 •	99.47 ± 0.16 ◦
optdigits	95.36 ± 0.21	95.83 ± 0.30 •	95.79 ± 0.20 •	95.90 ± 0.23 •	95.96 ± 0.26 •	96.11 ± 0.48 ◦	95.69 ± 0.65
ring	94.70 ± 0.19	93.83 ± 0.31 •	94.13 ± 0.26 •	93.93 ± 0.23 •	93.81 ± 0.25 •	94.09 ± 0.13 •	94.46 ± 0.29 •
sick	98.31 ± 0.25	94.01 ± 0.51 •	94.09 ± 0.51 •	93.78 ± 0.42 •	94.06 ± 0.43 •	98.54 ± 0.08 ◦	98.11 ± 0.44
spambase	94.34 ± 0.15	93.11 ± 0.27 •	93.17 ± 0.16 •	93.18 ± 0.20 •	93.14 ± 0.21 •	93.37 ± 0.13 •	91.93 ± 0.37 •
thyroid	99.77 ± 0.02	97.93 ± 0.26 •	97.63 ± 0.25 •	97.82 ± 0.44 •	97.76 ± 0.51 •	99.72 ± 0.03 •	99.62 ± 0.16 •
twonorm	96.71 ± 0.22	96.50 ± 0.20 •	96.55 ± 0.13 •	96.45 ± 0.21 •	96.69 ± 0.13	94.91 ± 0.39 •	95.45 ± 0.26 •
waveform	83.78 ± 0.45	83.45 ± 0.36 •	83.40 ± 0.43 •	83.47 ± 0.22 •	83.57 ± 0.33	82.18 ± 0.36 •	82.90 ± 0.81 •
wine	71.72 ± 0.97	40.55 ± 1.74 •	40.60 ± 1.07 •	40.86 ± 1.44 •	40.92 ± 1.37 •	67.46 ± 0.45 •	65.23 ± 1.81 •
Number of bests	14	1	0	1	1	3	6
Wins of ES-JSS		20.5	21	20.5	18.5	17.5	16

- Compared to known bagging ensemble methods, such as Bagging and RF, ensemble selection by ES-JSS was necessary and more efficient in most cases. It can not only make the ensemble more compact but also improve the performance.
- ES-JSS outperformed known boosting ensemble methods, such as XGBoost and GBDT, on more than half of the datasets. This

illustrates that our proposed ES-JSS is better than or comparable to known boosting ensemble methods in many cases.

These observations indicate that ES-JSS outperforms state-of-the-art technologies for both the ensemble and selective ensemble learning in most cases.

Table 4

Mean and standard deviation results for the accuracy of ES-JSS and four known ensemble methods. For each dataset, the best values are in bold font. '•' and '◦' respectively denote that ES-JSS was significantly better or worse than the given method using a t-test with a confidence level of 0.05. The 'Number of bests' row lists the numbers of datasets for which the best accuracy was obtained. The 'Wins of ES-JSS' row lists the numbers of datasets for which ES-JSS was significantly better than the corresponding method using the t-test (a tie was counted as 0.5 wins).

Datasets	ES-JSS	Bagging	RF	XGBoost	GBDT
adult	84.37 ± 0.10	83.95 ± 0.26 •	84.20 ± 0.30	85.50 ± 0.30 ◦	85.90 ± 0.29◦
australian	86.61 ± 1.33	86.71 ± 1.95	87.11 ± 1.42	87.23 ± 1.92	85.95 ± 1.28
cancer	98.46 ± 0.64	95.67 ± 1.30 •	96.32 ± 0.83 •	95.56 ± 1.27 •	94.50 ± 2.45 •
census	95.70 ± 0.02	95.53 ± 0.10 •	95.70 ± 0.08	95.78 ± 0.10◦	95.73 ± 0.08
coil2000	94.10 ± 0.10	92.69 ± 0.63 •	93.14 ± 0.68 •	92.43 ± 0.65 •	93.53 ± 0.49 •
cloumn	86.62 ± 2.38	80.90 ± 3.98 •	79.62 ± 3.17 •	80.00 ± 2.91 •	80.38 ± 3.02 •
credit	81.24 ± 0.13	80.55 ± 0.32 •	81.07 ± 0.41	81.02 ± 0.38	82.03 ± 0.40◦
crowd	91.94 ± 0.14	90.57 ± 0.68 •	91.76 ± 0.25 •	93.62 ± 0.33◦	92.37 ± 0.49 ◦
eeg	90.04 ± 0.33	87.81 ± 1.01 •	88.30 ± 0.72 •	89.44 ± 0.56 •	80.39 ± 0.65 •
fars	76.23 ± 0.06	75.78 ± 0.34 •	75.65 ± 0.32 •	76.81 ± 0.34 ◦	77.27 ± 0.28◦
flare	52.25 ± 0.81	46.62 ± 2.22 •	47.32 ± 2.97 •	47.22 ± 2.97 •	46.79 ± 2.35 •
FPS-5	79.09 ± 0.42	75.88 ± 1.72 •	82.20 ± 1.28◦	80.90 ± 1.27 ◦	80.27 ± 0.99 ◦
german	77.42 ± 1.62	73.16 ± 1.56 •	74.84 ± 1.93 •	71.72 ± 2.10 •	72.28 ± 2.63 •
letter	91.55 ± 0.18	90.16 ± 0.50 •	92.40 ± 0.32◦	91.41 ± 0.33	88.71 ± 0.50 •
magic	88.03 ± 0.06	86.66 ± 0.58 •	86.95 ± 0.49 •	86.91 ± 0.44 •	86.65 ± 0.44 •
market	31.53 ± 0.42	29.41 ± 1.33 •	29.95 ± 1.15 •	29.41 ± 1.19 •	32.41 ± 1.57
nursery	99.12 ± 0.10	98.26 ± 0.35 •	96.70 ± 0.36 •	99.46 ± 0.22◦	98.79 ± 0.31 •
optdigits	95.36 ± 0.21	95.21 ± 0.63	96.86 ± 0.39◦	96.14 ± 0.53 ◦	95.80 ± 0.67 ◦
ring	94.70 ± 0.19	93.10 ± 0.83 •	93.97 ± 0.53 •	95.17 ± 0.55◦	94.84 ± 0.44
sick	98.31 ± 0.25	96.80 ± 0.89 •	96.23 ± 0.78 •	97.01 ± 0.69 •	96.90 ± 0.36 •
spambase	94.34 ± 0.15	93.01 ± 0.89 •	94.44 ± 1.02	94.40 ± 0.98	94.13 ± 0.91
thyroid	99.77 ± 0.02	99.46 ± 0.14 •	99.15 ± 0.21 •	99.43 ± 0.17 •	99.26 ± 0.30 •
twonorm	96.71 ± 0.22	96.19 ± 0.63 •	96.88 ± 0.38	96.29 ± 0.45 •	96.46 ± 0.60
waveform	83.78 ± 0.45	82.28 ± 1.20 •	84.39 ± 1.00	83.76 ± 0.91	84.02 ± 1.23
wine	71.72 ± 0.97	65.41 ± 2.34 •	65.87 ± 2.40 •	63.75 ± 2.14 •	64.30 ± 2.20 •
Number of bests	10	0	6	5	4
Wins of ES-JSS		24	18.5	14.5	15.5

In the following, we use two standard unsupervised feature selection schemes, Sparse k -means and LW k -means, on our transformed data. Sparse k -mean and LW k -means are wrapper unsupervised selection methods, while ES-JSS is a filter unsupervised feature selection method. The following comparison, seen in Table 5, demonstrates whether wrapper unsupervised feature selection is suitable ensemble selection.

From Table 5, we have the following observations. 1) Ensemble selection based on Sparse k -means and LW k -means techniques works well. This demonstrates that it is feasible and effective by directly introducing the idea of unsupervised feature selection into ensemble selection. 2) Furtherly, we observe that ES-JSS outperforms Sparse k -means and LW k -means based ensemble selection. This shows that filter feature selection is a more suitable to ensemble selection than those wrapper methods. This is because the main goal of ES-JSS is to get a better ensemble selection result, whereas the goal of Spare k -means and LW k -means is a better and meaningful clustering result.

5.3. Sensitivity and Convergence

In this subsection, we consider the sensitivity of parameters and the convergence of ES-JSS. Here, we only report the results on six datasets, as shown in Figs. 3, 4, and 5. Similar observations can be obtained from other datasets.

The performance is relatively sensitive to the number of selected base learners, p . With increasing p , as shown in Fig. 3, the accuracy increases. Fig. 3 further illustrates that the trendline increases noticeably at the beginning and remains invariant or even decreases slightly when p is large. The optimal value of parameter p is data-dependent and remains an open problem.

The trendlines in Fig. 4 indicate that the number of clustering centers, c , has little impact on the performance of the selective

Table 5

Mean and standard deviation results for the accuracies of ES-JSS and two standard unsupervised feature selection. For each dataset, the best values are in bold font. '•' and '◦' respectively denote that ES-JSS was significantly better or worse than the given method using a t-test with a confidence level of 0.05. The 'Number of bests' row lists the numbers of datasets for which the best accuracy was obtained. The 'Wins of ES-JSS' row lists the numbers of datasets for which ES-JSS was significantly better than the corresponding method using the t-test (a tie was counted as 0.5 wins).

Datasets	ES-JSS	Sparse k -means	LW k -means
adult	84.37 ± 0.10	84.74 ± 0.19◦	84.55±0.28
Australian	86.61 ± 1.33	85.38±3.07	85.61±2.67
cancer-683	98.46 ± 0.64	97.40±2.24	97.87±1.49
census	95.70 ± 0.02	95.52±0.05•	95.54±0.05•
coil2000	94.10 ± 0.10	93.62±0.38•	93.4±0.41•
column_3c_weka	86.62 ± 2.38	83.90±4.17	83.90±3.96
credit	81.24 ± 0.13	80.45±0.15•	79.99±0.25•
crowd	91.94 ± 0.14	90.66±0.35•	91.05±0.62•
eeg	90.04 ± 0.33	87.81±0.57•	87.7±0.69•
fars	76.23 ± 0.06	76.20±0.19	76.18±0.18
flare	52.25 ± 0.81	51.74±3.27	51.48±1.29
FPS-5	79.09 ± 0.42	76.68±1.68•	76.07±1.4•
german	77.42 ± 1.62	74.60±3.20	73.71±3.26
letter	91.55 ± 0.18	90.42±0.72•	90.6±0.75•
magic	88.03 ± 0.06	86.71±0.51•	86.78±0.21
marketing	31.53 ± 0.42	30.24±0.70•	31.03±1.1
nursery	99.12 ± 0.10	98.86±0.26	98.76±0.22•
optdigits	95.36 ± 0.21	94.68±0.66•	94.57±0.74•
ring	94.70 ± 0.19	94.03±0.56•	94.1±0.91
sick	98.31 ± 0.25	97.78±0.39•	97.58±0.16•
spambase	94.34 ± 0.15	93.5±0.74•	93.35±0.77•
thyroid	99.77 ± 0.02	99.57±0.12•	99.57±0.14•
twonorm	96.71 ± 0.22	94.75±0.63•	94.79±0.6•
waveform	83.78 ± 0.45	82.89±0.76•	82.91±0.73•
winequality-red	71.72 ± 0.97	66.65±2.43•	67.46±1.72•
Number of bests	24	1	20
Wins of ES-JSS		20.5	20

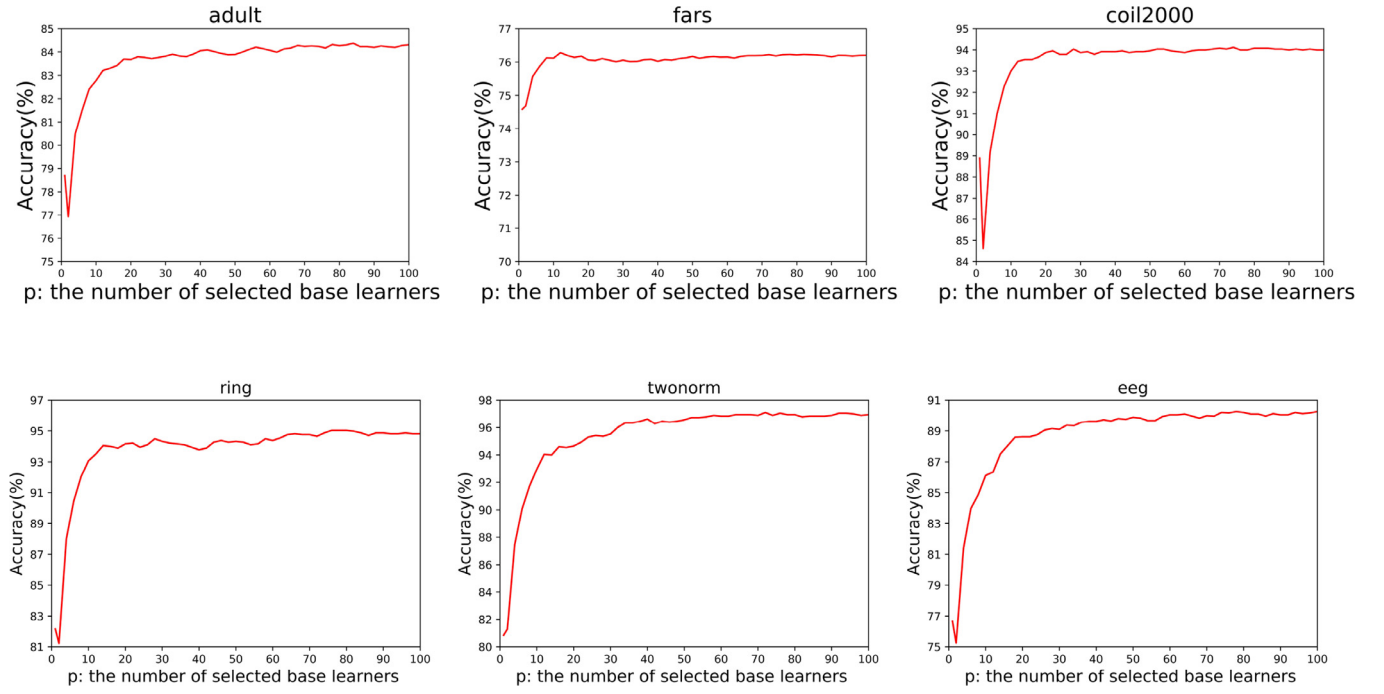


Fig. 3. Performance of ES-JSS with different p when $\lambda = 0.05$, $\theta = 2.0$, and $c = 2$.

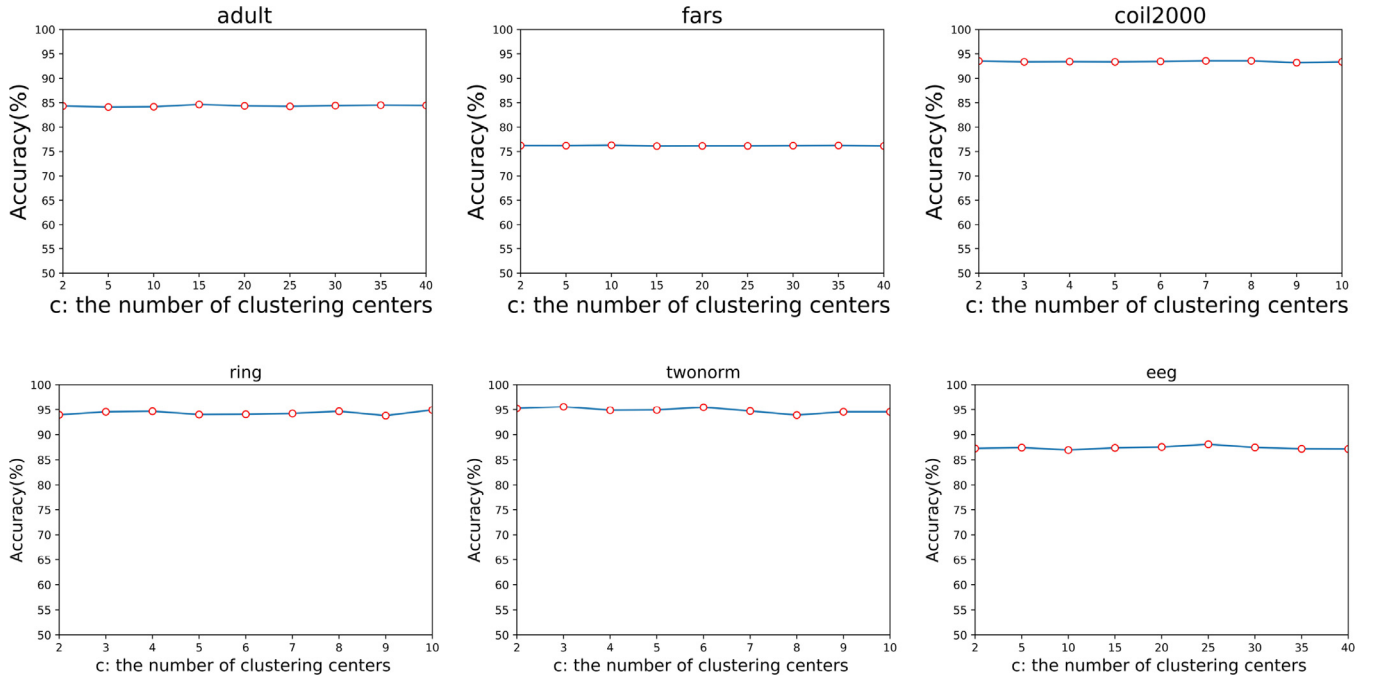


Fig. 4. Performance of ES-JSS with different c when $\lambda = 0.05$, $\theta = 2.0$, and $p = 20$.

ensemble. However, c affects the convergence speed, as shown in Fig. 5. This is because increasing c learns more sophisticated clustering structures, which does not affect ES-JSS's performance but takes more time to converge. Furthermore, Fig. 5 demonstrates that our proposed optimization algorithm converges after 20 iterations,

which shows that our proposed ES-JSS is effective and converges quickly.

In the following, we set $p = 20$ and experimentally demonstrate whether our proposed ES-JSS achieves the similar selective ensembles when three different types classifiers are adopted as base

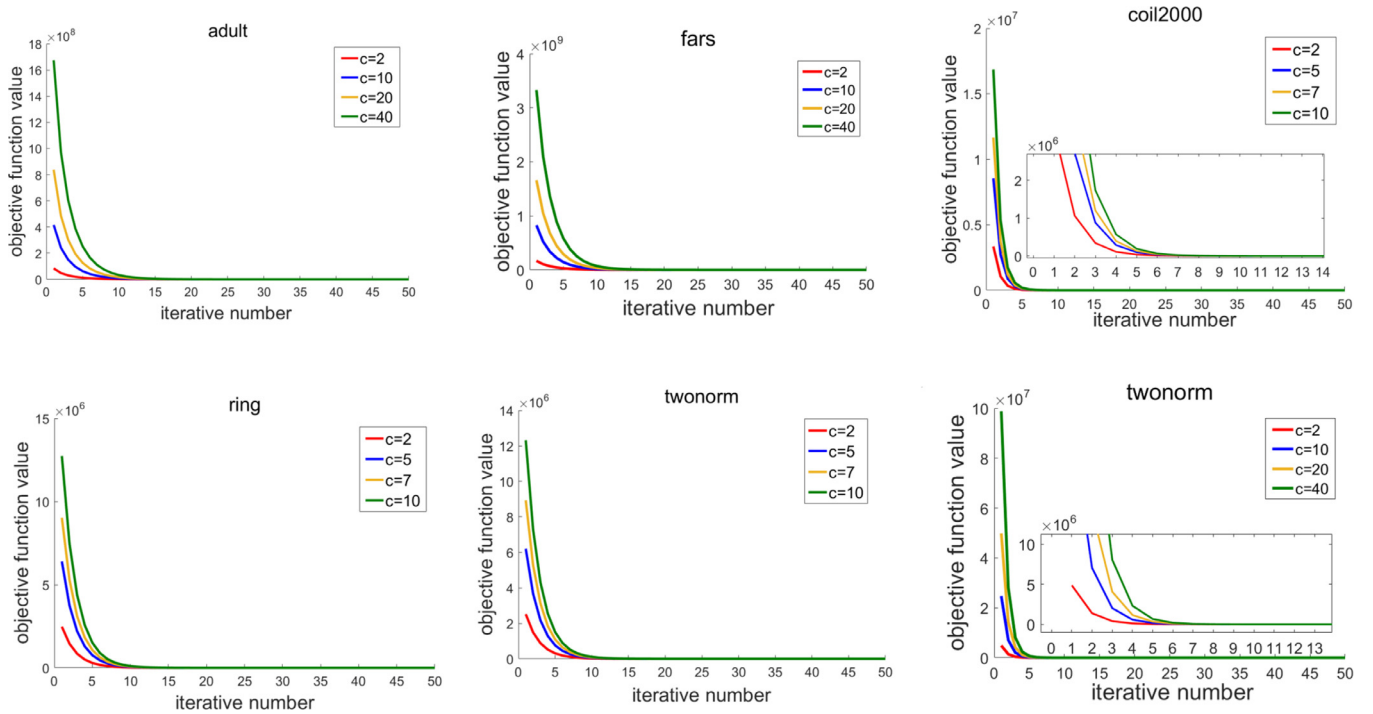


Fig. 5. Convergence curve of ES-JSS.

Table 6
Sensitivity to the types of base learners.

	Type of base learners	Selective Ensemble	Accuracy of selective ensemble	Accuracy of ensemble
ring	DT	6 14 23 27 35 36 40 41 42 44 53 54 58 68 70 73 83 86 88 94	94.20	95.60
	NN	6 14 23 27 35 36 40 41 44 53 54 58 68 70 73 76 83 86 88 94	60.64	62.50
	Bayes	6 14 23 27 35 36 40 41 44 53 54 58 68 70 73 83 86 88 89 94	97.71	97.65
adult	DT	9 16 22 27 30 31 32 37 41 42 44 49 50 56 68 74 76 80 87 94	84.21	84.92
	NN	9 16 22 27 30 31 32 37 41 42 44 49 50 56 68 74 76 80 87 94	82.80	82.83
	Bayes	9 16 22 27 30 31 32 37 41 42 44 49 50 56 68 74 76 80 87 94	80.46	80.45
fars	DT	2 4 5 7 15 18 27 32 33 43 44 48 52 56 74 84 88 92 96 99	76.15	76.22
	NN	2 4 5 7 15 18 27 32 33 43 44 48 52 56 74 84 88 92 96 99	61.10	61.00
	Bayes	2 4 5 15 18 27 28 30 32 33 43 44 48 52 56 69 74 84 92 96	71.56	72.02
coil2000	DT	1 9 16 18 23 32 33 35 44 51 56 61 63 74 78 79 82 89 92 94	92.96	93.45
	NN	1 9 16 18 23 32 33 35 44 51 56 61 63 74 78 79 82 89 92 94	93.67	93.66
	Bayes	1 9 12 16 18 23 32 33 35 44 56 61 63 74 78 79 82 89 92 94	11.27	11.30
twonorm	DT	0 5 8 17 18 30 32 35 50 51 55 63 67 70 81 82 83 84 86 99	95.41	96.77
	NN	0 5 8 17 18 30 32 35 50 63 67 70 72 81 82 83 84 86 95 99	94.37	93.78
	Bayes	0 5 8 11 17 18 30 32 35 50 63 67 70 81 82 83 84 86 95 99	97.76	97.87
eeg	DT	8 10 11 13 21 30 41 44 49 52 53 54 58 64 70 75 79 86 92 99	85.92	87.75
	NN	8 10 11 13 21 24 30 41 44 49 52 54 58 64 70 75 79 86 92 99	55.36	55.36
	Bayes	8 11 13 21 24 30 41 44 49 53 54 58 64 70 75 79 86 92 96 99	61.78	61.70

learner of ensemble. And then the selective ensembles of ES-JSS on different types of base learners, such as Neural Networks, Decision trees and Bayesian Network, are presented in Table 6. We get several observations from Table 6. First, the results of ensemble selection on different types of base learner are almost the same, indicated by the 'Selective ensemble' column. This shows that our ES-JSS is robust to the type of base learners. Second, although the types of base learners are different, the prediction loss after ensemble selection is tiny, or even there is no loss. This demonstrates that in our proposed ES-JSS the type of base learner does not affect the performance of ensemble selection.

6. Conclusions

Considering that previously proposed static ensemble selection models are not flexible, we proposed a robust static ensemble se-

lection method by integrating spectral clustering and structural sparsity into a joint framework, named ES-JSS. A key operation of our method is transforming the labeled original data to an unlabeled prediction space, which makes unsupervised learning for ensemble selection, i.e., ES-JSS, feasible and effective. One main strength of the proposed ES-JSS is its less sensitivity to the type of base learners. This demonstrates that the key information for static ensemble selection lies on the prediction space of the base learners; rather than on the class labels of the original data, just as demonstrated by our extensive experiments on 25 datasets. However, ES-JSS requires that the prediction of base learner should be a probabilistic value; only then can it transform the original data into correct prediction space.

To our best knowledge, this is the first effective unsupervised framework for ensemble selection. In the near future, we plan to incorporate more powerful self-supervised learning techniques for

ensemble selection on the unlabeled prediction space. It would be interesting to evaluate whether they evidently deliver a comparable or better performance when instances in the unlabeled prediction space are contrasted as positive or negative samples for feature selection.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

In this appendix, we consider the sensitivity of parameters of ES-JSS, λ and θ . Here, we only report the results on six datasets, as shown in Figs. A.6 and A.7. Similar observations can be obtained from other datasets.

The experimental results presented in Figs. A.6 and A.7 demonstrate that the prediction performance fluctuates in an insignificant range that is often barely distinguishable. This shows that the learned weight matrix with row sparsity can capture the discriminative base learners regardless of whether λ and θ are large or small. The above observations show that our method is not significantly sensitive to λ and θ with wide ranges.

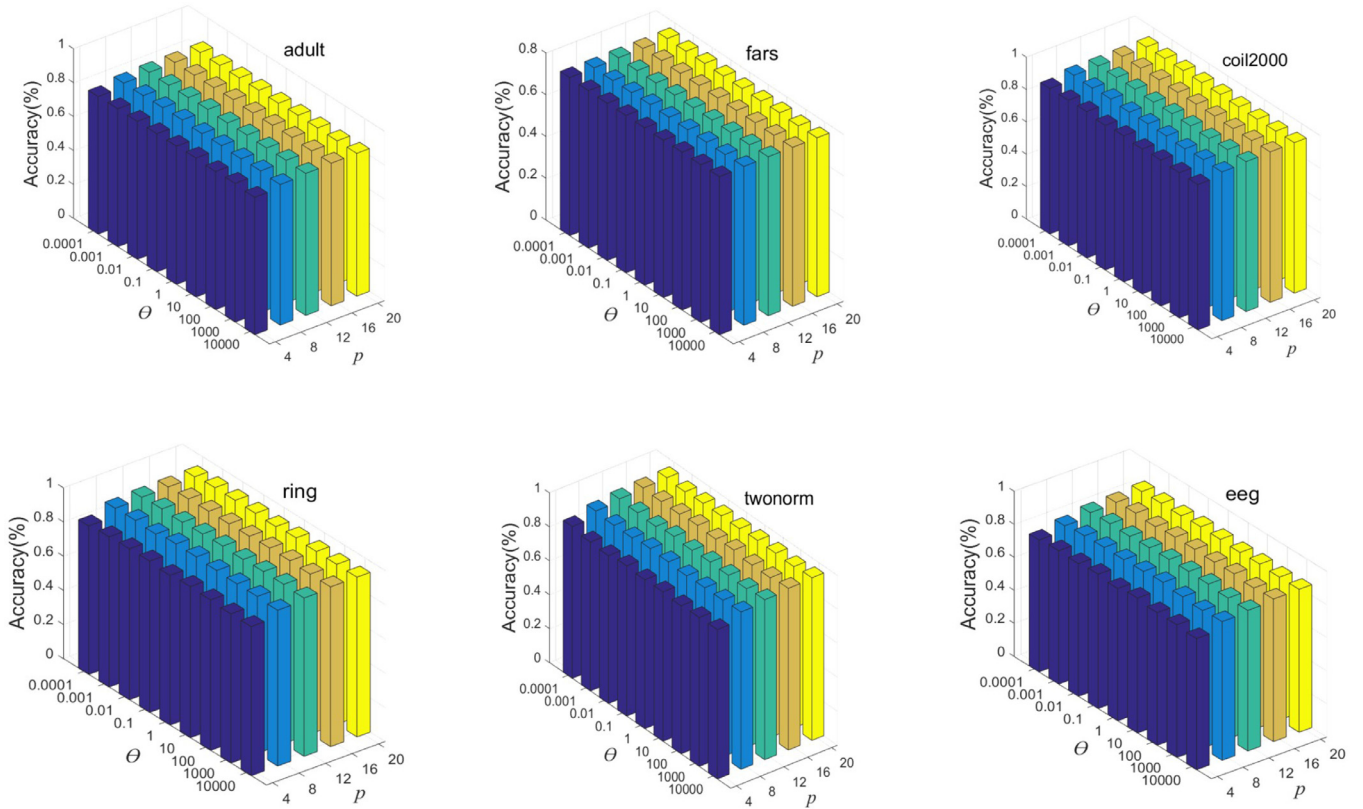


Fig. A6. Performance of ES-JSS with different θ and p values when $\lambda = 0.05$ and $c = 2$.

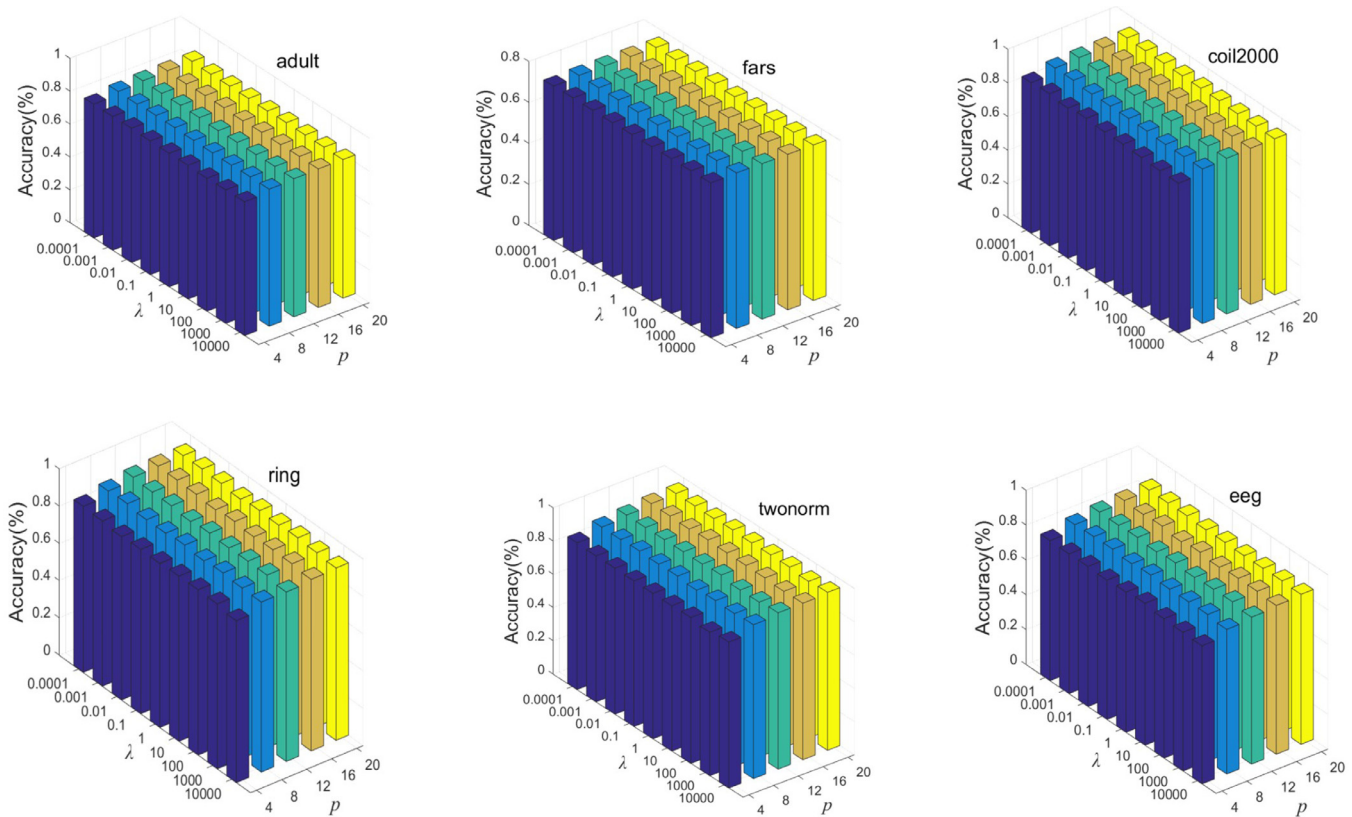


Fig. A7. Performance of ES-JSS with different λ and p values when $\theta = 2.0$ and $c = 2$.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.patcog.2021.108061](https://doi.org/10.1016/j.patcog.2021.108061).

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